

X3D Agent Suite

Ontology-Driven Multi-Agent Architecture for Standards-Based 3D Authoring

Executive Summary

The X3D Agent Suite decomposes 3D authoring into cooperating agent skills responsible for ontology, perspective, semantics, validation, and publishing.

Architecture

Ontology=WHAT exists; Perspective=WHO experiences it; Semantics=HOW entities relate; Representation=HOW encoded.

Architecture Diagram

User Intent -> Planner -> Router -> Ontology -> Perspective -> Semantics -> Representation
-> Validation -> Publishing

Perspective Architect

Authoritative source of gravity, scale, embodiment, navigation and viewpoints.

Canonical World Frame

Gravity=-Y, Up=+Y, Right=+X, Forward=-Z, Right-handed.

Repository Structure

Includes source code, ontology, SHACL, schemas, examples, tests and docs.

Release Information

Release: v1.0.0 Perspective Architect reference implementation.

Layer	Question	Example Skill
Ontology	What exists?	Ontology Reasoner
Perspective	Who experiences?	Perspective Architect
Semantics	How related?	Semantic Reasoner

Representation

How encoded?

Graphics Author